

MENTAL HEALTH IN CRISIS

Changing Patterns in U.S.
Emergency Departments

Exploratory Data Analysis | CDC National Syndromic Surveillance Program

January 2022 - May 2026

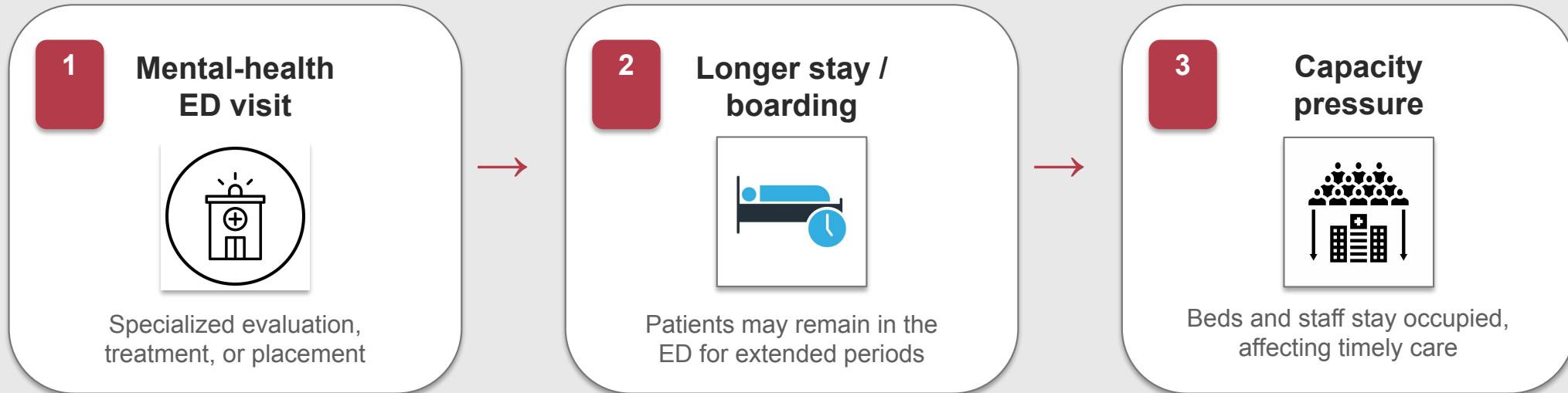
TruBridge Data Analytics Project Presentation

Research question

How did ED visit rates for major mental health conditions change, and which conditions exhibited the largest observed changes?

Mental-health demand can tie up ED capacity beyond the initial visit.

Why do condition-level changes matter for hospital operational planning?



Planning Value: Condition-specific patterns can guide staffing, psychiatric capacity, resource allocation, and surveillance.

Research Question

How did ED visit rates for major mental health conditions change, and which conditions exhibited the largest observed changes?

ANALYSIS OBJECTIVE

Identify which mental-health conditions are changing, without claiming that any one condition causes ED crowding.

CDC Dataset scope

Primary analysis uses the overall U.S. population subset of a CDC NSSP Mental Health Emergency Department Visit Rate Dataset; demographic cuts are supporting context.

10,591

rows in the raw dataset

6

individual mental health conditions

Jan 2022 - May 2026

analysis start to end

53

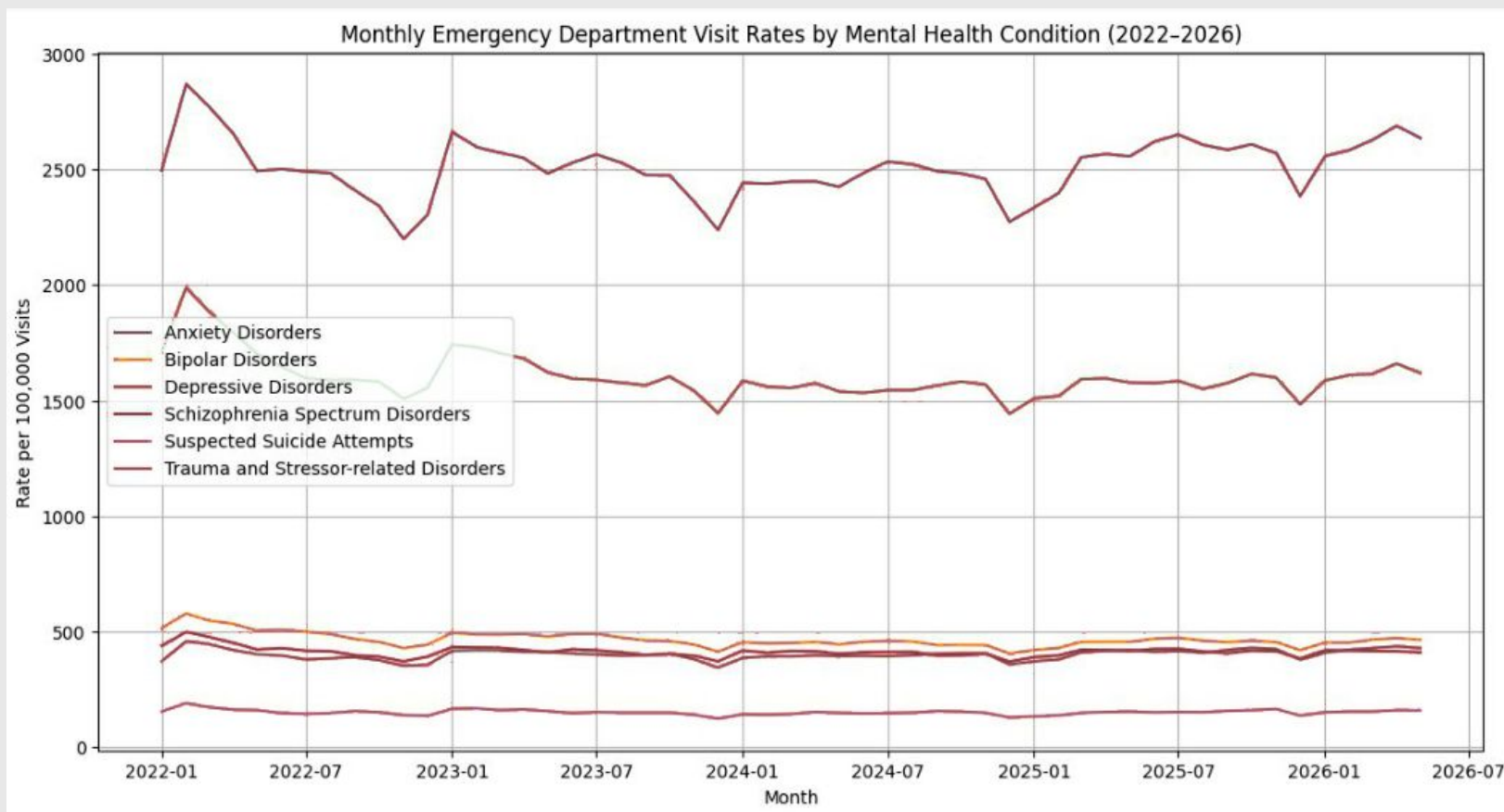
of months analyzed

Dataset Preparation Choices

- Converted month_end to datetime.
- Evaluated potential outliers with IQR; many reflected real demographic differences.
- Restricted the national trend analysis to demographics_type = “Total.”
- Removed aggregate “All” rows from demographic subsets.
- Excluded “Any Mental Health” from condition-level analysis because it is an aggregate category.

Anxiety dominates visit rates, but conditions move differently over time.

Monthly rates show distinct trajectories rather than one uniform mental-health pattern.



What to notice

2,713.99

mean visit rate for Anxiety Disorders - highest of the six conditions

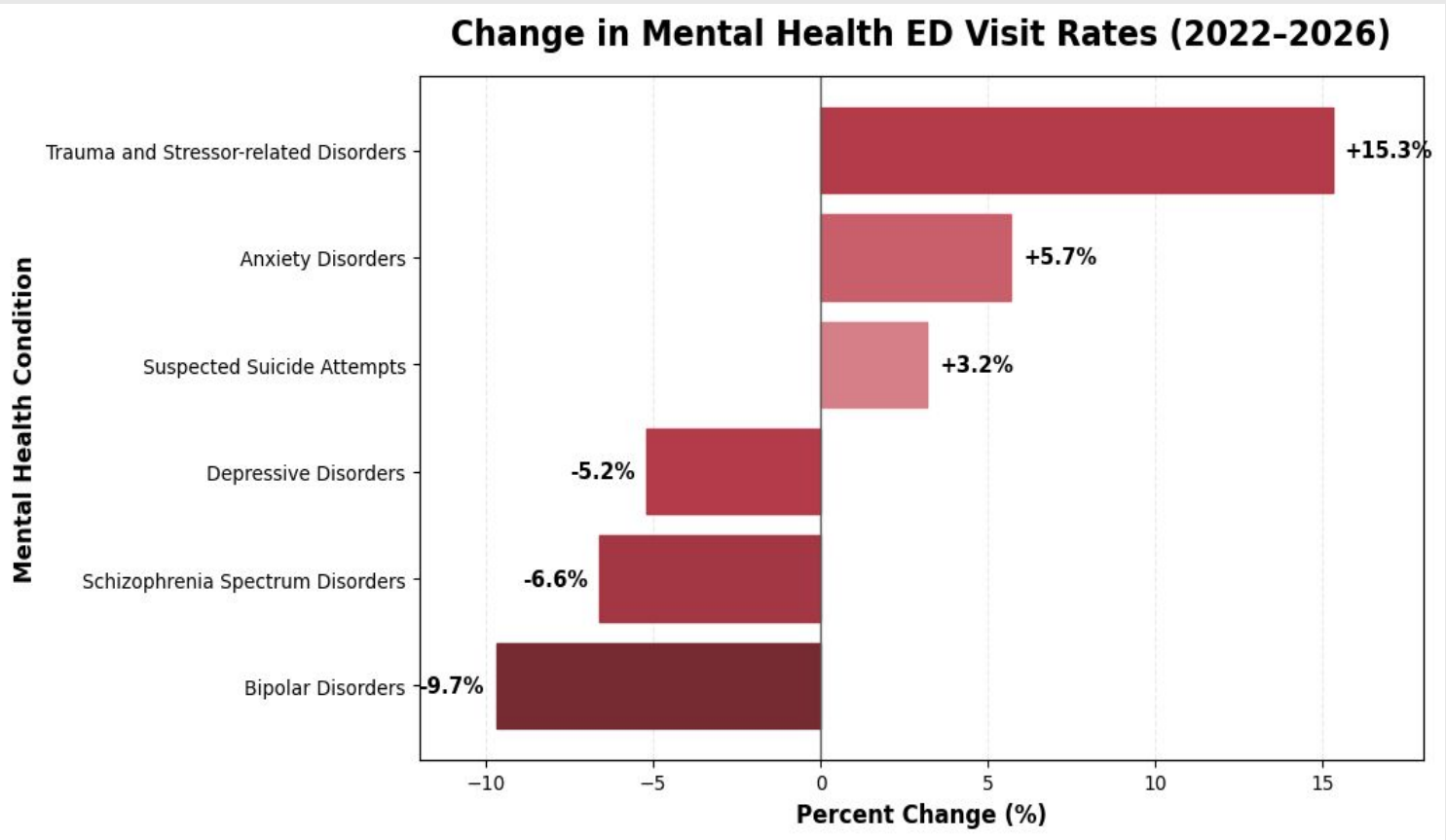
151.76

mean visit rate for Suspected Suicide Attempts - lowest of the six

*** The monthly variation makes endpoint-only comparisons incomplete.**

Trauma-related ED visit rates show the largest endpoint increase.

From January 2022 to May 2026, Trauma and Stressor-related Disorders visit rates have increased by 15.3%.



ENDPOINT CHANGE

+15.3%

Trauma & Stressor-related Disorders

- +5.7% Anxiety
- +3.2% Suspected Suicide Attempts
- 5.2% Depressive
- 6.6% Schizophrenia Spectrum
- 9.7% Bipolar

Endpoint growth describes the difference between two months - not the full trajectory.

Demographic context highlights groups with higher pooled mean visit rates.

These summaries combine all six conditions and are descriptive context - not condition-specific causal comparisons.

1,237.12

Age 12-17: highest pooled mean rate

933.83

Age 65+: lowest pooled mean rate

1,122.98

Female highest pooled mean rate

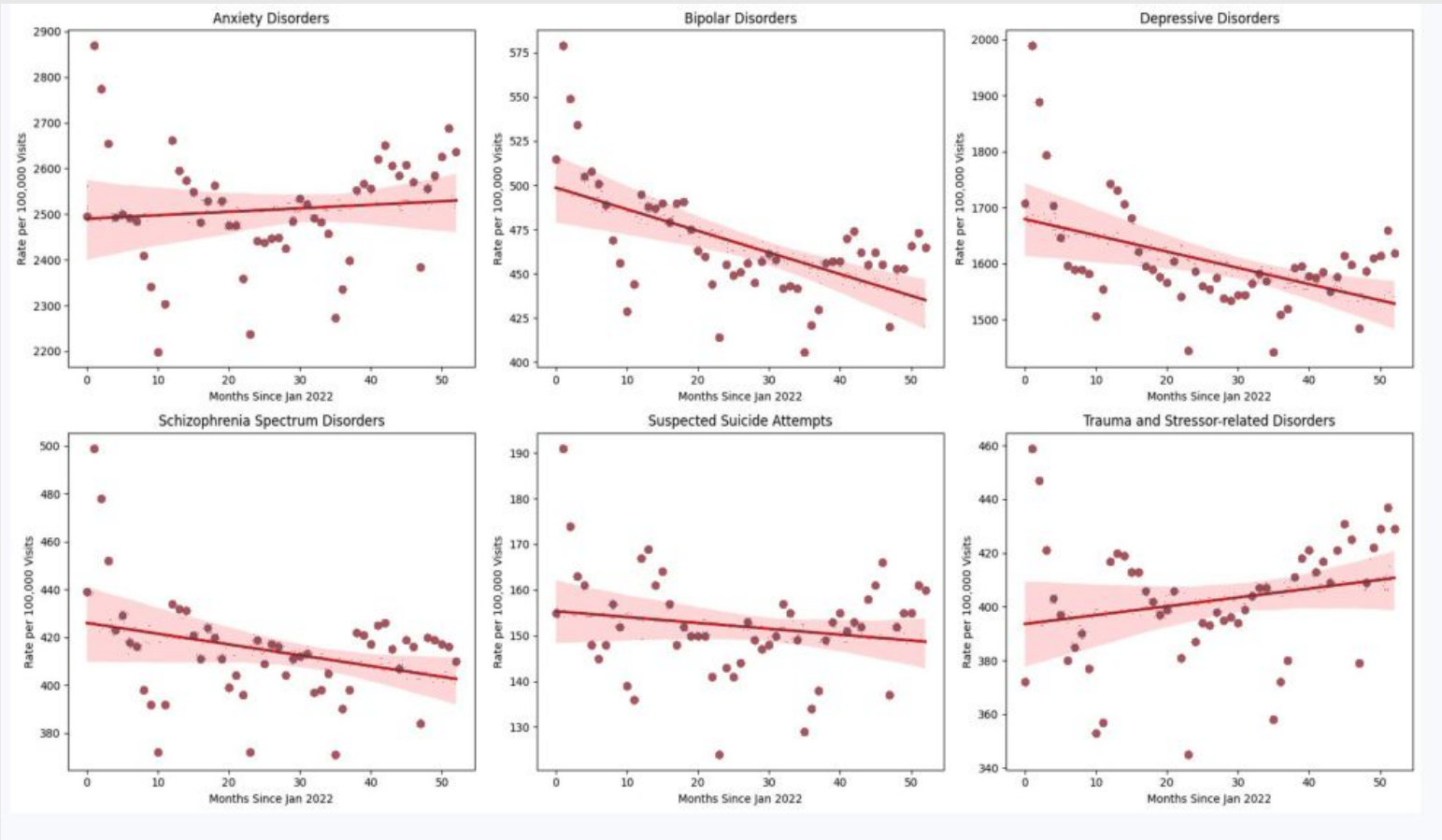
Race / Ethnicity Rates

White observations had the highest pooled mean rate (1,226.52), followed by AI/AN (1,191.92); Asian observations had the lowest (608.25).

These results motivate deeper condition-by-demographic analysis, not to infer why the differences exist.

Endpoint growth does not equal sustained upward trends.

Regression tests show that the apparent increases in trauma and anxiety are not statistically significant linear trends.



Regression takeaway

No significant upward linear trend

Anxiety: $p = 0.5034$

Trauma: $p = 0.1140$

Suspected suicide attempts:
 $p = 0.2080$

Significant downward linear trends

Depressive: $p = 0.0004$

Bipolar: $p < 0.001$

Schizophrenia spectrum: $p = 0.0214$

Depressive Disorders had the steepest estimated decline: -2.8966 rate points per month.

The strongest planning signal is divergence, not universal growth

Different measures answer different questions: endpoints flag change; regression tests consistency.

Monitor for possible growth

- **Trauma +15.3% endpoint**
- **Anxiety +5.7% endpoint**

Neither showed sufficient evidence of a consistent upward linear trend across all monthly observations.

Consistent downward signal

- **Depressive**
- **Bipolar**
- **Schizophrenia Spectrum**

All three had statistically significant negative slopes over the observed period.

Action: monitor emerging increases and investigate what drives them

The analysis supports targeted follow-up rather than a claim that ED mental-health demand is rising uniformly.

1 Monitor



Continue tracking trauma- and anxiety-related visit rates as additional 2026 observations become available.

2 Segment



Break trends down by condition within age, sex, and race/ethnicity to identify which populations may be driving changes.

3 Model



Evaluate seasonality, nonlinear patterns, abrupt changes, and autocorrelation rather than relying only on a simple linear trend.

Key takeaway: Changing composition matters more than a single growth story.

Trauma and stressor-related disorders had the largest endpoint increase (+15.3%), but the full monthly series did not establish a statistically significant upward linear trend.

For healthcare planning, endpoint changes can flag conditions to watch, while trend models help determine whether those changes are sustained.

Thank you for listening!

Questions?

Source: CDC National Syndromic Surveillance Program (NSSP)